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KENYATTA UNIVERSITY

UNIVERSITY EXAMINATIONS 2019/2020

FIRST SEMESTER EXAMINATION FOR THE DEGREE OF BACHELOR OF SCIENCE IN ELECTRICAL AND ELECTRONIC ENGINEERING

EEE 202: COMPUTER PROGRAMMING II

DATE: Friday 31st January 2020 **TIME: 11.00a.m -1.00p.m**

INSTRUCTIONS

- There are **FIVE** questions in this exam. Answer Question **ONE** and any other **TWO** Questions
- Clearly show your solutions. Neatness and organization in your work is very important
- No electronic device such as calculators and phones may be used in this exam
- Your phone **MUST** be completely switched off and kept away
- You are not allowed to share any exam materials with other students
- **DO NOT** place your solutions in a manner that other students can see them

Question One [30 Marks]

a) Say you've been hired by Microsoft to write a compiler for C*, a new language that's just like C++, except that Microsoft can add whatever new language features it wants without the trouble of going through a standards body. One of your co-workers proposes a solution to the problem of C++ exceptions causing memory leaks. She suggests that when an exception is thrown in C*, the language is smart enough to call delete or delete[] (as appropriate) on every pointer it encounters as it unwinds the stack. In other words, stack-based objects are destructed just like in C++, but pointers are deleted. Describe briefly why this is a bad idea. **[4 Marks]**

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b) Write a C++ program to compute x^n using a recursive function. Your program should consist of the main function from which the user is requested for inputs, x and n, and an additional function, *f_recursive()*, that implements x^n. A part of the algorithm for x^n is as shown below: [4 Marks]

If n is equal to 0 then return 1;

If n is equal to 1 then return x;

If n is even then ....

If n is odd then .....

- instance of - data structure - member function - declared in class and created when an object is instantiated

c) Explain the terms object, class, methods, and instant variables as used in C++ programming.

[4 Marks]

d) Given:

string my_string;

Briefly explain the main difference between the following two statements: [4 Marks]

cin << my_string;

getline(cin, my_string);

e) An enumerated type declares an optional type name and a set of zero or more identifiers that can be used as values of the type. Each enumerator is a constant whose type is the enumeration. Illustrate how the enumerated type is used in C++ programs. [4 Marks]

f) A storage class defines the scope (visibility) and life-time of variables and/or functions within a C++ Program. These specifiers precede the type that they modify. Briefly explain the working of any FOUR storage classes that can be used in a C++ Program. [4 Marks]

g) Using an appropriate example distinguish between syntax and semantics of a C++ program.

[3 Marks]

h) One of the most important concepts in object-oriented programming is that of *inheritance*. Inheritance allows us to define a class in terms of another class, which makes it easier to create and maintain an application. How useful is the concept of inheritance to a C++ programmer? [3 Marks] f the

Question Two [20 Marks]

a) Write a function named *"eliminate_duplicates"* that takes an array of integers in random order and eliminates all the duplicate integers in the array. The function should take two arguments: (i) an array of integers; (ii) an integer that tells the number of cells in the array. The function should not return a value, but if any duplicate integers are eliminated, then the function should change the value of the argument that was passed to it so that the new value tells the number of distinct integers in the array. **[6 Marks]**

b) Please find out what is wrong with the *if statement* listed below, and rewrite it to fix that logical error. (Hint: what if someone's GPA is 1.0). Your new *if statement* should contain nested *if-else-if statements.* **[5 Marks]**

```
if ((gpa >= 2.3) && (gpa < 3.0))
   cout << "Your GPA is not bad but you should improve it.";
if ((gpa >= 3) && (gpa <= 3.7))
   cout << "You got a pretty good GPA";
else
   cout << "Congratulations. You made the Dean's list";
```

c) Using a suitable program code distinguish between a constructor and a destructor. Your program should include a piece of code that produces the following output: **[5 Marks]**

Object is being created
<any object related output as may be defined in your code>
Object is being deleted

d) C++ polymorphism means that a call to a member function will cause a different function to be executed depending on the type of object that invokes the function. Using an example C++ code segment explain and illustrate how this is achieved. **[4 Marks]**

Question Three [20 Marks]

a) Write a C/C++ program that prompts the user to enter the year and month, and displays the number of days in the month. For example, if the user entered month 2 and year 2000, the program should display that February 2000 has 29 days. If the user entered month 3 and year 2005, the program should display that March 2005 has 31 days. **[6 Marks]**

b) Write a C++ function that takes in a 2-dimensional integer array with 10 rows and 5 columns as inputs and returns the maximum value of that array and its location within the array as reference parameters. **[6 Marks]**

c) The main purpose of C++ programming is to add object orientation to the C programming language and *classes* are the central feature of C++ that supports object-oriented programming. Using the code segment below explain the definition and working of *C++ classes*. **[4 Marks]**

```
class Box {
  public:
    double length;   // Length of a box
    double breadth;  // Breadth of a box
    double height;   // Height of a box
};
```

d) "A *do..while*loop is almost the same as a while loop except that the loop body is guaranteed to execute at least once." Determine the validity of this statement and justify your answer using appropriate code segment examples. **[4 Marks]**

Question Four [20 Marks]

a) Using the example program codes given below explain the differences between the *break* and *continue* statements. Given the input 1.1, 2.2, 5.5, 4.4, -3.4, -45.5, 34.5, -4.2, -1000, and 12, provide the outputs for each piece of code. **[7 Marks]**

1.1, 2.2, 5.5, 4.4, -3.4, -45.5

34.5, -4.2 -1000 12

```c
# include <stdio.h>
int main()
{
      int i;
      double number, sum = 0.0;

      for(i=1; i <= 10; ++i)
      {
            printf("Enter a n%d: ",i)
            scanf("%lf",&number);

            // If user enters negativ
            if(number < 0.0)
            {
                  continue;
            }

            sum += number; // sum =
      }

      printf("Sum = %.2lf",sum);

      return 0;
}
```

```c
# include <stdio.h>
int main()
{
      int i;
      double number, sum = 0.0;

      for(i=1; i <= 10; ++i)
      {
            printf("Enter a n%d: ",i);
            scanf("%lf",&number);

            // If user enters negative
            if(number < 0.0)
            {
                  break;
            }

            sum += number; // sum = s
      }

      printf("Sum = %.2lf",sum);

      return 0;
}
```

1.1+0 = 1.1 + 2.2 + 5.5 + 4.4 + 34.5 + 12

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b) Write a C++ program which opens a file in reading and writing mode. After writing information entered by the user to the file named myfile.txt the program should read information from the file and output it onto the screen. **[7 Marks]**

c) C++ arrays allow you to define variables that combine several data items of the same kind. Explain any TWO methods that are used in the realization of this. **[6 Marks]**

Question Five [20 Marks]

a) Write a complete program that contains a main function and at least two additional functions to compute the area and perimeter of a square. The program should ask the user to enter a double number corresponding to the side length of the square, and display the area and perimeter of the square on the screen. **[7 Marks]**

b) A person is described as having a name, sex, age, an employer, salary, etc. Write a C++ Program for the structure person, assign data to members of the structure variable and display the data. **[6 Marks]**

c) The Fibonacci numbers are defined:

$$fibonacci\ (0) = 1$$

$$fibonacci\ (1) = 1$$

$$fibonacci\ (n) = fibonacci\ (n-1) + fibonacci\ (n-2);$$

Write a recursive C++ function to compute the Fibonacci numbers. **[4 Marks]**

d) Consider the following C function, intended to receive a string as an argument and return a reversed string as a result. (The strlen function returns the number of characters in the string).

```
#include <string.h>
#include <stdlib.h>
char *reverse(char *input)
        { int len = strlen(input);
        char output[len];
        for (int i = 0; i < len; i++)
        { output[len - i - 1] = input[i]; }
        return output;
        }
```


Identify THREE bugs in this function. **[3 Marks]**

c) Write the declaration for a class called dog that contains two data members: a string called breed and an int called age. (Don't include any member functions).

(5 marks)

d) What are the three pieces usually necessary when using "while" loop in c++ programming?

condition

(3 marks)

e) What is Standard Template Library

(2 marks)

Question Four

a) Write a code segment that multiplies each element of an array by 2, storing it back in the array.

(3 marks)

b) Define inheritance and explain the 4 types of inheritance.

(9 marks)

c) List and explain the class member's visibility

(6 marks)

d) What is operator overloading?

(2 marks)

Question Five

a) Create a class called time that has separate int member data fro hours, minutes, and seconds. One constructor should initialize this data to 0, and another should initialize it to fixed values. Another member function should display it, in 11:59:59 format. The final member function should add two objects of type time passed as arguments. A main () program should create two initialized time objects (should they be const?) and one that isn't initialized. Then it should add the two initialized valued together, leaving the result in the third time variable. Finally it should display the value of this third variable. Make appropriate member functions const.

(10 marks)

c) A phone number, such as (212) 767-8900, can be thought of as having area code (212), the exchange (767), and the number (8900). Write a program that uses a structure to store these three parts of a phone number separately. Call the structure phone. Create two structure variables of type phone. Initialize one, and have the user input a number for the other one. Then display both numbers. The interchange might look like this:

Enter your area code, exchange, and number : 415 555 1212

My number is (212)767-8900

Your number is (415) 555-1212

(10 marks)

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```
public:
        A( );
        void get( );
        void put( );
};
class B:protected A
{
        int c,d;
    protected:
        int e,f;
        void get2( );
    public:
        B( );
        void put2( );
};
class C: private B
{
        int p;
    protected:
        int q;
        void get3( );
    public:
        void show3( );
};
```

int size; length, width.
int area
{
Printf("Enter double number")
scanf("%" &length)
Printf("The

ii) A function isPositive that takes as input a double number and returns the number if the number is positive, and zero otherwise.

d) What does it mean for a language to be strongly typed? Statically typed? What may prevent a programming language from being strongly typed? [5 Marks]

e) Explain the relationship between overloading and dynamic dispatch. [2 Marks]

Question Five [20 Marks]

a) Write a complete program that contains a main function and at least two additional functions to compute the area and perimeter of a square. The program should ask the user to enter a double number corresponding to the side length of the square, and display the area and perimeter of the square on the screen. [6marks]

b) What is "polymorphism"? How is it implemented in C++? [3 Marks]

c) What are "function overloading" and "function overriding" in C++? Are they the same? If not, explain how they are different. [3 Marks]

d) Consider the C++ declaration below and answer the following questions:

i) Name all the member functions which are accessible by the objects of class C. [2 Marks]

ii) Name all the protected members of class B [2 Marks]

iii) Name the base class and derived class of class B. [2 Marks]

iv) Name all the data members which are accessible from member functions of class C [2 Marks]

```
class A
{
   void any();
   protected:
        int a,b;
        void proc();
```

f- positive (int)

f- function - positive(int)

f- input number(double)

int no;

Print if (input

KENYATTA UNIVERSITY

UNIVERSITY EXAMINATIONS 2014/2015

FIRST SEMESTER EXAMINATION FOR THE DEGREE OF BACHELOR OF SCIENCE

EEE 202: COMPUTER PROGRAMMING II

DATE: THURSDAY 11TH DECEMBER 2014 TIME: 2.00 P.M. – 4.00 P.M.

INSTRUCTIONS

Attempt question One and any other Two Questions

Question One is Compulsory and carries 30 marks.

The remaining questions carry 20 marks each.

Question One

a) Outline the characteristics of object oriented programming. (8 marks)

b) Write the statements that will display the following on the screen

i) the character 'x' (1 mark)

ii) the name james (1 mark)

iii) the number 509 (1 mark)

c) Name the two exceptions to the rule that the compiler ignores the whitespace (2 marks)

d) Write a do lop that display the number from 100 to 110 (5 marks)

e) Write the process of programming in an object-oriented language? (3 marks)

f) Give any four advantages of OOP (3 marks)

g) What are the basic concepts of OOP (6 marks)

(5 1.

Question Two

(1 mai

a) Define the following terms

(1 mark)

i) Encapsulation

(1 mark)

ii) Polymorphism

(1 mark)

iii) Preprocessor directive

(1 mark)

iv) Classes

v) Objects

(4 marks)

b) Write a program that generates the following table

2010	135
2012	7290
2013	11300
2014	16200

Use a single statement for all output

c) What header file must you include with your source file to use cout and cin?

(1 mark)

(5 marks)

d) Write five special characteristics of constructor.

(3 marks)

e) List and indicate the three logical operators and their effect in c++

(2 marks)

f) What is unnamed namespaces?

Question Three

a) Write a class definition that creates a class called leverage with one private data member, crowbar, of type int and one public function whose declaration is void pry() (5 marks)

b) For a two dimensional array of type float, called flarr, write a statement that declares the array and initializes the first subarray to 52, 27, 83; the second to 94, 73, 49; and the third to 3,6,1. (5 marks)

```
i = 3;
j = 4;
Mystery (y,x,j,i);
cout << x << y << i << j;
```

c) Explain the following terms in the context of object-oriented programming languages

i) A class [2 Marks]

ii) Instance of an object [2 Marks]

iii) List any TWO objected-oriented features of C++ [2 Marks]

d) A positive integer n is said to be prime (or, "a prime") if and only if n is greater than 1 and is divisible only by 1 and n. For example, the integers 17 and 29 are prime, but 1 and 38 are not prime. Write a function named "is_prime" that takes a positive integer argument and returns as its value the integer 1 if the argument is prime and returns the integer 0 otherwise. [4marks]

Question Four [20 Marks]

a) What is data encapsulation? How is it implemented in C++? Your friend announces that he implemented data encapsulation in C, which is not an OOP language. Do you agree with him or not? Explain clearly. [4marks]

b) The Fibonacci numbers are defined:

int fibonacci (numb)
int (numb==0)
return 1.
if numb==1
return 1.

fibonacci (0) = 1

fibonacci (1) = 1

fibonacci (n) = fibonacci (n–1) + fibonacci (n–2);

Write a recursive C++ function to compute the Fibonacci numbers. [3 Marks]

c) Write some C++ program segments that solves the indicated tasks (you do not have to write a complete program or be concerned about "good" output; a small code segment will be sufficient).

i) A program that gets a double number from the user, decides whether that number is positive, negative, or zero and display its decision on the screen. [3 Marks]

double

if ==0

else

cout << "Congratulations. You made the Dean's list";

c) The Greatest Common Divisor (GCD) of a pair of integers m and n (not both zero) is the positive integer d that is a divisor of both m and n. Write a function named "g_c_d" that takes two positive integer arguments and returns as its value the greatest common divisor of those two integers. If the function is passed an argument that is not positive (i.e., greater than zero), then the function should return the value 0 as a sentinel value to indicate that an error occurred. [5marks]

d) Since exception handling is not necessary in a language, explain why most modern languages have it. [3 Marks]

e) Provide a detailed comparison of structured programming languages and object-oriented programming languages. [3 Marks]

Question Three [20 Marks]

a) Explain static type-checking giving details of how it relates to dynamic scoping. Is it possible to do static type checking with dynamic scoping? Why, or why not? Give an example. [6 Marks]

b) Consider the following function:

```
void Mystery(double x, double y, int i, int j)
{
    x = y;
    y = x;
    i = 2*i;
    j = 2*j;
}
```

What is the output when the function *Mystery* is used as in the code segment below? [4 Marks]

```
double x = 10.0;
double y = 20.0;
int i = 30;
int j = 40;
Mystery (x,y,i,j);
cout << x << y << i << j;
x = 1.0;
y = 2.0;
```

Describe unary, binary, and ternary operators giving an example of each in C++ language. [2 Marks]

) If you have a const pointer in C++ can you change the object it points to? Explain. [4 Marks]

) Define a class REPORT with the following specification.

Private:

Adno	4 digit admission number
Name	20 characters
Marks	an array of floating point values
Averge	average marks obtained
Getavg()	to compute the average obtained in five subjects

Public:

Readinfo()	funtion to accept values for adno,name,marks and invoke the function getavg()
Displayinfo()	function to display all data members on the screen.

h) Describe the advantages and disadvantages associated with the basic data types *int* and *double*. [3 Marks]

i) Differentiate among public, protected and private members of a class. [3 Marks]

j) Write a small piece of code that defines an array A of 10 doubles and sets all of them to 1.0 [3 Marks]

Question Two [20 Marks]

a) What is aliasing? Give examples of constructs in C++ that may give rise to aliasing. Explain why aliasing is harmful. [4 Marks]

b) Please find out what is wrong with the *if statement* listed below, and rewrite it to fix that logical error. (Hint: what if someone's GPA is 1.0). Your new *if statement* should contain nested *if-else-if statements.* [5marks]

```
if ((gpa >= 2.3) && (gpa < 3.0))
  cout << "Your GPA is not bad but you should improve it.";
if ((gpa >= 3) && (gpa <= 3.7))
  cout << "You got a pretty good GPA";
```

KENYATTA UNIVERSITY

UNIVERSITY EXAMINATIONS 2015/2016

FIRST SEMESTER EXAMINATION FOR THE DEGREE OF BACHELOR OF SCIENCE (ELECTRICAL ENGINEERING)

EEE 202: COMPUTER PROGRAMMING II

DATE: Wednesday 9th December 2015 TIME: 8.00a.m-10.00a.m

INSTRUCTIONS:

- There are FIVE questions in this exam. Answer Question ONE and any other TWO Questions
- Clearly show your solutions. Neatness and organization in your work is very important
- No electronic device such as calculators and phones may be used in this exam
- Your phone MUST be completely switched off and kept away
- You are not allowed to share any exam materials with other students
- DONOT place your solutions in a manner that other students can see them

Question One [30 Marks]

a) Explain the distinction between decisions that are bound statically (at compile time) and those that are bound dynamically (at run time). [3 Marks]

b) Explain the differences between a compiled and an interpreted language giving details of where C++ belongs. [3 Marks]

c) Distinguish between pass-by-reference and pass-by-value? [3 Marks]

d) Using an appropriate example distinguish between syntax and semantics of a C++ program? [3 Marks]

Determine the output of the following code fragments. Marks]

i)
```
int x = 1, y = 0;
if (x > 0 && y < 0) {
x = y = 23; }
cout << x << " " << y << endl;
```
ii)
```
int x = 1, y = 0;
if (x > 0 || y < 0) {
x = y = 23; }
cout << x << " " << y << endl;
```
iii)
```
x = 10; y = 40;
if (x >= 10) {
if (y < 40) {
y++;
} else {
y--; }
cout << x << " " << y << endl;
```
iv)
```
x = 10; y = 40;
if (x >= 10) {
if (y < 40) {
y++;
}
} else {
y--; }
cout << x << " " << y << endl;
```

```
x=10; y=40;
if (x>=10)
{
if (y<40)
{
y++;
}
else
{
y--;
}
```


11. Using a suitable program code distinguish between a constructor and a destructor. Your program should include a piece of code that consists of parameterized constructors and destructors clearly showing the execution of the constructor, a class member function, and the destructor. **[10 Marks]**

12. Write a C++ program which opens a file in reading and writing mode. After writing marks entered by the user to a file named marks.txt, the program reads the marks from the file, increments the marks by 10, and writes it back to the file. **[10 Marks]**

13. Consider the following C++ declaration and answer the questions given below:

```
class A {
    int num1, num2;
    void any();
  protected:
    int a, b;
    void proc();
  public:
    A();
    void get();
    void put();
};

class B: protected A
{
    int c, d;
  protected:
    int e, f;
    void get2();
  public:
    B();
    void put2()
};

class C: private B, protected A
{
    int p, num3, num4
  protected:
    int q;
    void get3();
  public:
    void show3();
};
```

i) Name all the member functions which are accessible by the objects of class C. **[4 Marks]**
ii) Name all the protected members of class B **[5 Marks]**
iii) Name all the data members which are accessible from member functions of class C. **[4 Marks]**
iv) Name the base class and derived class of class B. **[2 Marks]**